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funkyheatmap: Visualising data frames with mixed data types

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DOI: 10.xxxxxx/draft

Software

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Editor: Julia Romanowska

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Submitted: 14 December 2024
Published: unpublished

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Summary

The {funkyheatmap} package offers a flexible and user-friendly solution for visualising data frames containing a mixture of categorical, proportional, and text-based data. It simplifies the creation of informative and visually appealing heatmaps while providing extensive customization options to tailor the output. This tool is especially valuable in research settings for summarising and communicating complex results, such as those encountered in benchmarking studies.

The package is available on CRAN, PyPI and npm. For a showcase of examples and an overview of the different implementations, visit the project website funkyheatmap.github.io.

Statement of need

Data visualisation is fundamental to exploratory data analysis and communicating findings. While powerful tools like ggplot2 (Wickham, 2009), Matplotlib (Hunter, 2007) and D3.js (Bostock et al., 2011) exist, they often require complex scripting to generate comprehensive visualisations for data frames containing a mix of data types. {funkyheatmap} addresses this challenge by:

- **Seamless Handling of Mixed Data**: Automates the selection of appropriate visualisations (rectangles, bars, pie charts, text) based on data types.
- **Customization**: Provides granular control over colours, groupings, geometries, and annotations for tailored results.
- **Accessibility**: Offers a simplified interface for basic use and detailed documentation for advanced customization.

{funkyheatmap} has proven its utility in benchmarking studies within single-cell omics (Li et al., 2023; Luecken et al., 2022, 2024; Saelens et al., 2019; Sang-Aram et al., 2024; Yan & Sun,

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2023), but its applications extend to diverse fields where visualisation of mixed data types is needed.

Functionality

Figure 1 showcases the functionality of {funkyheatmap}, namely:

- **Diverse Geometries**: Supports a range of geometries (rectangles, bars, pie charts, text, images) to effectively represent different data types.
- **Hierarchical Categorical Grouping**: Facilitates the organisation of rows and columns into semantic groups with distinct colour palettes.

Besides this, the package includes comprehensive documentation, vignettes, and a test suite.

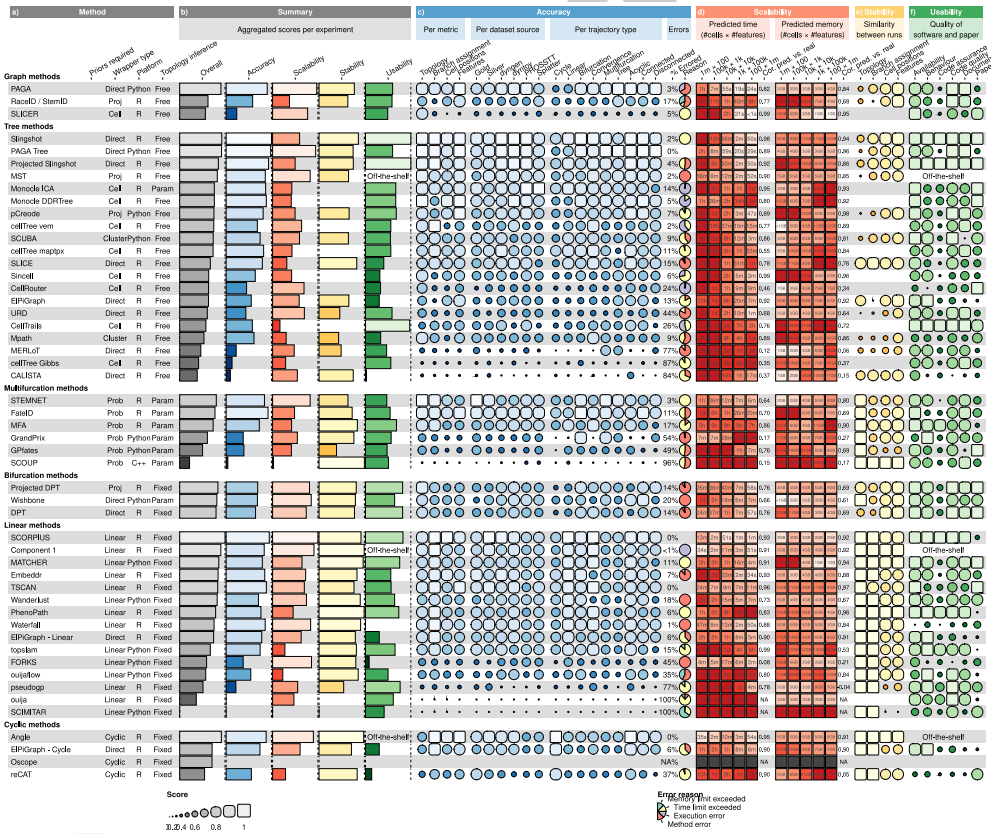

Figure 1: An example of a {funkyheatmap} visualisation using data from a benchmarking study of trajectory inference methods (Saelens et al., 2019).

See the following Table for more information regarding the recommended geom for different types of data.

Data type	Example	Recommended geom
Numerical data	Scores from 0 to 1	funkyrect
Aggregated data	The mean of scores	bar
Measurement data	3MB or 4h	rect + text overlay
Categorical data	R or Python	text or image
Proportional data	80% success, 10% OOM, 10% failed	pie

The table presents the suggested visualisation methods (geoms) based on the data type of the columns. These recommendations provide a starting point for users to select the most appropriate visual representation for their specific data.

Example usage

In order to produce a {funkyheatmap} visualisation, you need to provide the data in the form of a dataframe, which also must contain a column named `id`. If you provide no other information, a basic visualisation will be provided, but customization is possible by provinding additional information, such as a `column_info` dataframe which details how the columns in the dataframe get translated into different geoms, or a `row_groups` dataframe which allows you to group rows in the visualisation.

As an example, the following code produces Figure 2.

```
library(funkyheatmap)
library(dplyr, warn.conflicts = FALSE)
library(tibble, warn.conflicts = FALSE)

data("mtcars")

data <- mtcars %>%
  rownames_to_column("id") %>%
  arrange(desc(mpg))

column_info <- tribble(
  ~id,     ~group,        ~name,                        ~geom,        ~palette,
  "id",    "",            "",                           "text",       NA,
  "mpg",   "overall",     "Miles / gallon",             "bar",        "palette1",
  "cyl",   "overall",     "Number of cylinders",        "bar",        "palette2",
  "disp",  "group1",      "Displacement (cu.in.)",      "funkyrect",  "palette1",
  "hp",    "group1",      "Gross horsepower",           "funkyrect",  "palette1",
  "drat",  "group1",      "Rear axle ratio",            "funkyrect",  "palette1",
  "wt",    "group1",      "Weight (1000 lbs)",          "funkyrect",  "palette1",
  "qsec",  "group2",      "1/4 mile time",              "circle",     "palette2",
  "vs",    "group2",      "Engine",                     "circle",     "palette2",
  "am",    "group2",      "Transmission",               "circle",     "palette2",
  "gear",  "group2",      "# Forward gears",            "circle",     "palette2",
  "carb",  "group2",      "# Carburetors",              "circle",     "palette2",
)

funky_heatmap(
  data,
  column_info = column_info,
  position_args = position_arguments(expand_xmax = 4)
)
```

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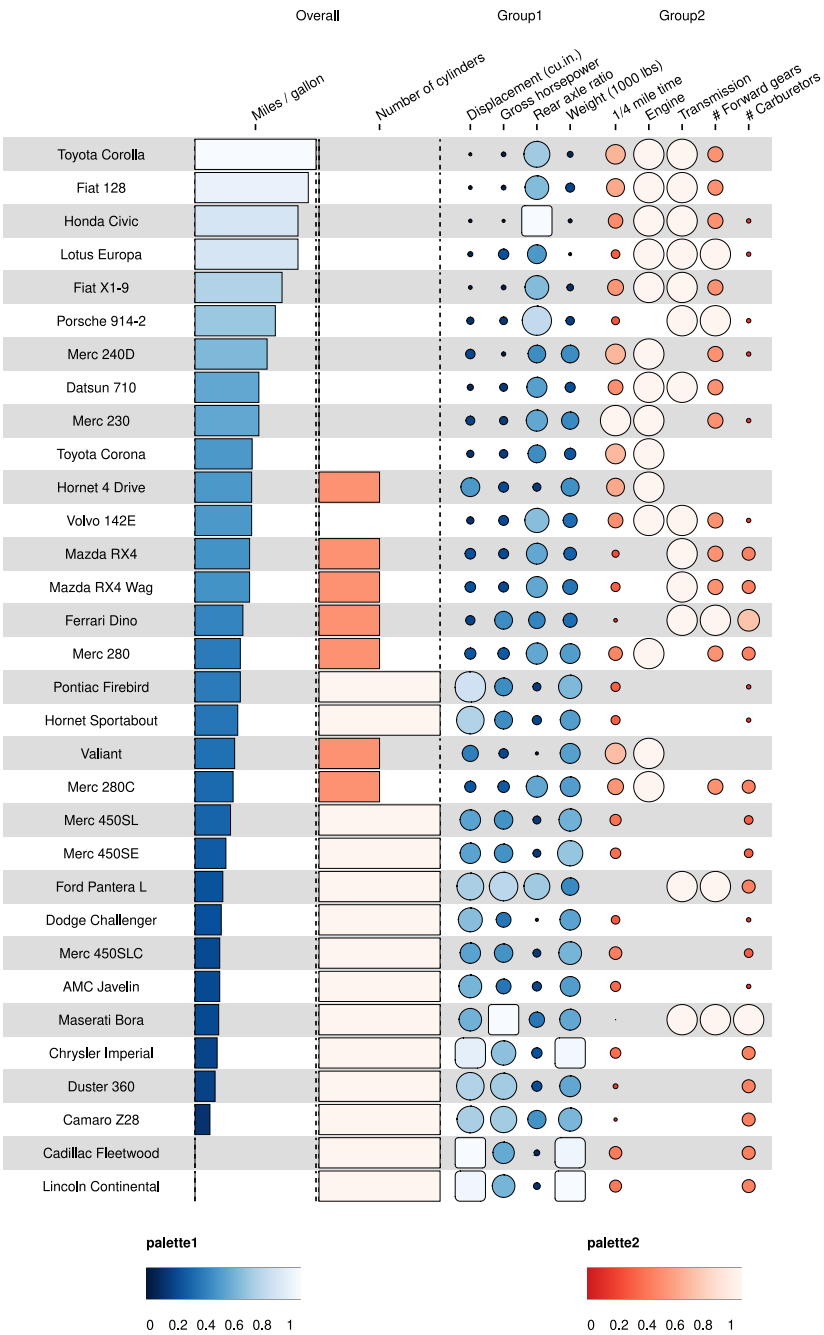

Figure 2: An example of a `{funkyheatmap}` visualisation using the mtcars dataset.

Conclusion

`{funkyheatmap}` streamlines the creation of publication-quality visualisation for mixed data types, empowering researchers and data scientists to communicate their results effectively. The development of `funkyheatmappy` (Python) and `funkyheatmapjs` (JavaScript) will further expand the accessibility and functionality of this visualisation solution.

Acknowledgements

L.Z. acknowledges support from the Bavarian Ministry of Science and the Arts in the framework of the Bavarian Research Association "ForInter" (Interaction of human brain cells). L.D. acknowledges support from the Research Foundation - Flanders (FWO) (1SF3822N). Y.S. acknowledges support from the Flemish Government under the "Onderzoeksprogramma Artificiële Intelligentie (AI) Vlaanderen" program. A.C. and Y.S. are funded by an FWO TBM grant (Research Foundation – Flanders, T000119N), and Ghent University Special Research Fund (BOF18-GOA-024). W.S. was supported by a Marie Skłodowska-Curie fellowship (101028476).

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